

Vulnerability Assessment Report

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System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 20XX to August 20XX. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

Consider the following questions to help you write:

- *How is the database server valuable to the business?*
- *Why is it important for the business to secure the data on the server?*
- *How might the server impact the business if it were disabled?*

Thanks to this database, the company finds new customers, however, as the data is public there are several threats such as the exfiltration of data and the possible deletion/alteration of it. Because of this, it is highly important to secure the data on the server. If not, there's a high chance the competitors will start stealing the potential customers of the database, resulting in a loss of credibility for the company.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Competitor	Obtain sensitive information via exfiltration	1	3	3
Hacker	Conduct "man-in-the-middle" attacks	2	3	6
System administrator	Alter/Delete critical information	3	3	9

Approach

As the database is public, a competitor could take a look into it and use the data on it for their own purpose , after this, they could create a plan to overlap the company and take advantage of it. Besides this, if a hacker gets into the network he could conduct a man in the middle and divert the data requests made by employees to sell it to the competitors. Another threat to consider is that an employee could alter or delete information by mistake, making it catastrophic if there is no backup.

Remediation Strategy

In order to remediate this, we are making the following suggestions:

- 1 - The principle of least privilege should be applied, in order to accomplish this, the database should be private and an implementation of an authentication factor (MFA, passwords, etc)is needed.
- 2 - Data encryption should be done with TLS not SSL as this method is already deprecated and is known for having critical vulnerabilities. This would help prevent hacker attacks.
- 3 - Daily backups should be made on a separate storage (Cloud or physical) to ensure the continuity of businesses in case of an Alter/delete mistake or a hardware issue.